

Employee Name: Jeswanth Saravanamuniyandi

Employee Id: 12568

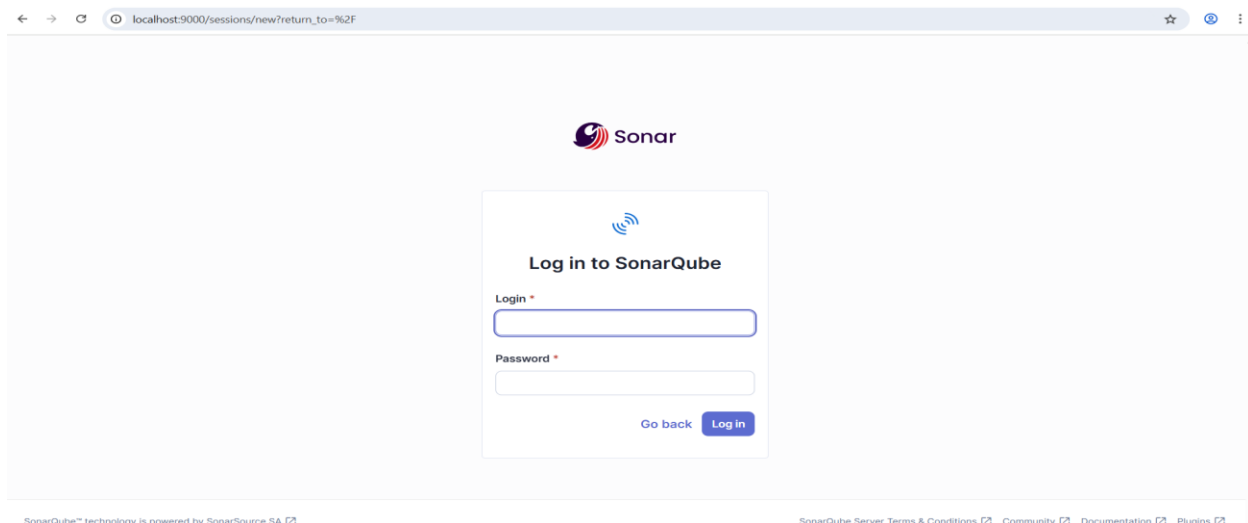
Sonar Report:

Step 1: Start Sonar for its Directory

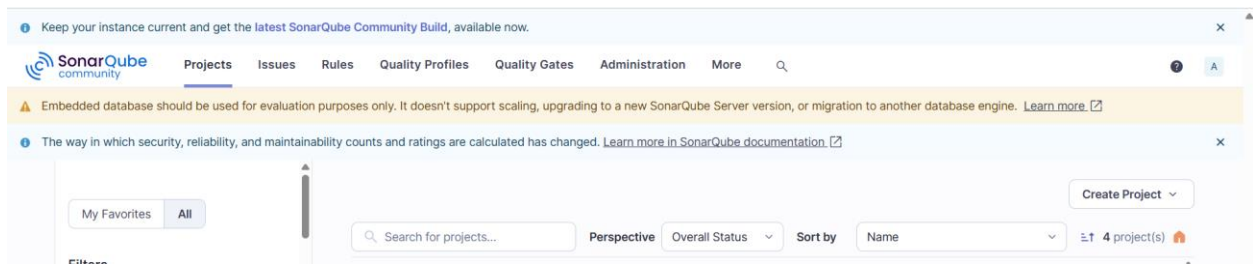
```
C:\WINDOWS\system32\cmd. X + v
Starting SonarQube...
2025.09.20 15:14:25 INFO app[[o.s.a.AppFileSystem] Cleaning or creating temp directory D:\Tools\sonarqube-25.1.0.10212\temp
2025.09.20 15:14:25 INFO app[[o.s.a.es.EsSettings] Elasticsearch listening on [HTTP: 127.0.0.1:9001, TCP: 127.0.0.1:{}]
2025.09.20 15:14:25 INFO app[[o.s.a.ProcessLauncherImpl] Launch process[ELASTICSEARCH] from [D:\Tools\sonarqube-25.1.0.10212\elasticsearch]: C:\Program Files\Java\jdk-17\bin\java -Xms4m -Xmx64m -XX:+UseSerialGC -Dcli.name=server -Dcli.script=.bin/elasticsearch -Dcli.libs=lib/tools/server-cli -Des.path.home=D:\Tools\sonarqube-25.1.0.10212\elasticsearch -Des.path.conf=D:\Tools\sonarqube-25.1.0.10212\temp\conf\es -Des.distribution.type=tar -cp D:\Tools\sonarqube-25.1.0.10212\elasticsearch\lib\*;D:\Tools\sonarqube-25.1.0.10212\elasticsearch\lib\cli-launcher\* org.elasticsearch.launcher.CliToolLauncher
2025.09.20 15:14:25 INFO app[[o.s.a.SchedulerImpl] Waiting for Elasticsearch to be up and running
Standard Commons Logging discovery in action with spring-jcl: please remove commons-logging.jar from classpath in order to avoid potential conflicts
2025.09.20 15:15:40 INFO app[[o.s.a.SchedulerImpl] Process[es] is up
2025.09.20 15:15:40 INFO app[[o.s.a.ProcessLauncherImpl] Launch process[WEB_SERVER] from [D:\Tools\sonarqube-25.1.0.10212]: C:\Program Files\Java\jdk-17\bin\java -Djava.awt.headless=true -Dfile.encoding=UTF-8 -Djava.io.tmpdir=D:\Tools\sonarqube-25.1.0.10212\temp -XX:-OmitStackTraceInFastThrow --add-opens=java.base/java.util=ALL-UNNAMED --add-opens=java.base/java.lang=ALL-UNNAMED --add-opens=java.base/java.io=ALL-UNNAMED --add-opens=java.rmi/sun.rmi.transport=ALL-UNNAMED --add-exports=java.base/jdk.internal.ref=ALL-UNNAMED --add-opens=java.base/java.nio=ALL-UNNAMED --add-opens=java.base/sun.nio.ch=ALL-UNNAMED --add-opens=java.management/sun.management=ALL-UNNAMED --add-opens=jdk.management/com.sun.management.internal=ALL-UNNAMED -Xmx512m -Xms128m -XX:+HeapDumpOnOutOfMemoryError -Dhttp.nonProxyHosts=localhost|127.*|:::1 -cp .\lib\sonar-application-25.1.0.10212.jar;D:\Tools\sonarqube-25.1.0.10212\lib\jdbc\h2\h2-2.3.232.jar org.sonar.server.app.WebServer D:\Tools\sonarqube-25.1.0.10212\temp\sq-process14821159197235500362properties
WARNING: A terminally deprecated method in java.lang.System has been called
WARNING: System::setSecurityManager has been called by org.sonar.process.PluginSecurityManager (file:/D:/Tools/sonarqube-25.1.0.10212/lib/sonar-application-25.1.0.10212.jar)
WARNING: Please consider reporting this to the maintainers of org.sonar.process.PluginSecurityManager
WARNING: System::setSecurityManager will be removed in a future release
Standard Commons Logging discovery in action with spring-jcl: please remove commons-logging.jar from classpath in order
```

Login:

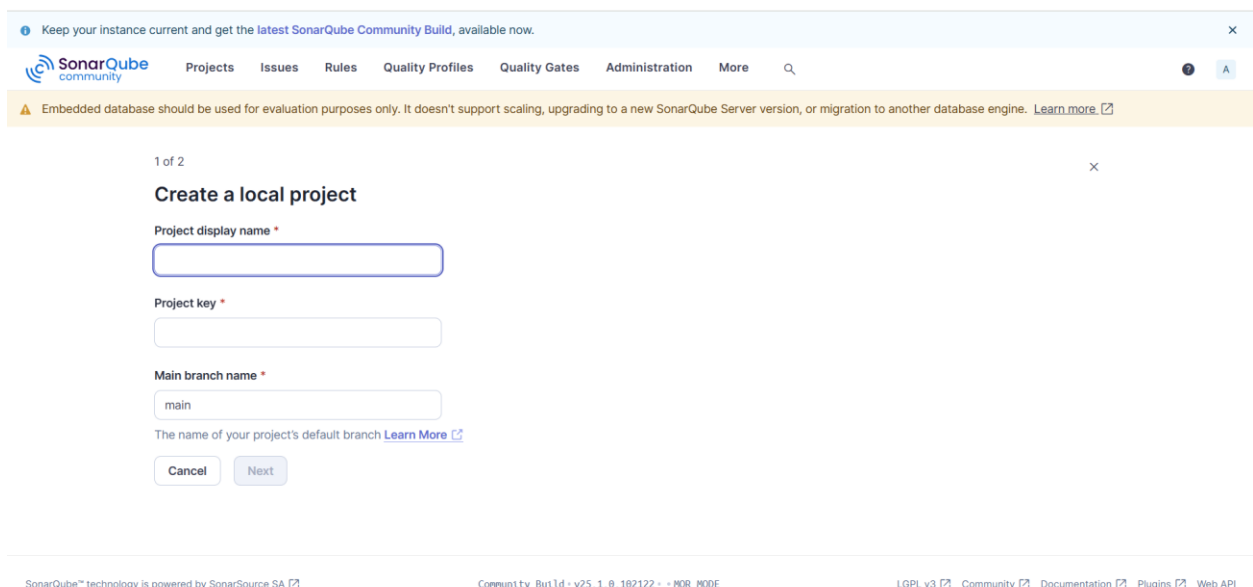
Go to: Localhost:9000



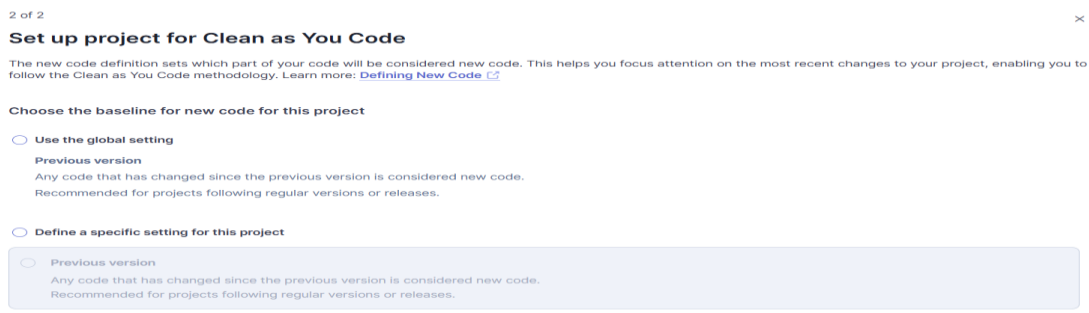
Step 3: Click “Create Project” in the Dashboard:



Step 4: Create a Local project and fill the details:



Step 5: Then setting up the project. Choose, ”use the global Setting” option for default version Setting.



And Click “Create Project”

Step 5: Select the Analysis Method as “Locally”

Step 6: Generate token for analyzing the project

1 Provide a token

Analyze "SonarAssessment": `sqp_93b74155d82503260f39c234d302235c1e0fedda` 

The token is used to identify you when an analysis is performed. If it has been compromised, you can revoke it at any point in time in your [user account](#).

Continue

2 Run analysis on your project

Step 7: Click “Continue” and Select your Project Details

2 Run analysis on your project

What option best describes your project?

☐ Maven ☐ Gradle ☒ .NET ☐ Other (for JS, TS, Go, Python, PHP, ...)

Which framework do you use?

☒ .NET Core ☐ .NET Framework

Step 8: Install the global Sonar scanner tool:

```
PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS COMMENTS
PS D:\Code Quality\ProductManager> dotnet tool install --global dotnet-sonarscanner
Tool 'dotnet-sonarscanner' is already installed.
PS D:\Code Quality\ProductManager> |
```

Step 9: Run the Sonarqube Analysis in your Project:

```

PS D:\Code Quality\ProductManager> dotnet sonarscanner begin /k:"SonarAssessment" /d:sonar.host.url="http://localhost:9000" /d:sonar.token="sqp_93b74155d82503260f39c234d302235c1e0fedda"
SonarScanner for MSBuild 10.3
Using the .NET Core version of the Scanner for MSBuild
Pre-processing started.
Preparing working directories...
15:32:44.568 Updating build integration targets...
15:32:44.743 Using SonarQube v25.1.0.102122.
15:32:44.885 Fetching analysis configuration settings...
15:32:46.561 Provisioning analyzer assemblies for cs...
15:32:46.562 Installing required Roslyn analyzers...
15:32:46.562 Processing plugin: csharp version 10.4.0.108396

```

Step 10: Build your Project:

```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS COMMENTS
15:32:46.561 Provisioning analyzer assemblies for cs...
15:32:46.562 Installing required Roslyn analyzers...
15:32:46.562 Processing plugin: csharp version 10.4.0.108396
15:32:47.843 Provisioning analyzer assemblies for vbnet...
15:32:47.843 Installing required Roslyn analyzers...
15:32:47.844 Processing plugin: vbnet version 10.4.0.108396
15:32:48.059 Incremental PR analysis: Base branch parameter was not provided.
15:32:48.06 Cache data is empty. A full analysis will be performed.
15:32:48.098 Pre-processing succeeded.
PS D:\Code Quality\ProductManager> dotnet build
Restore complete (0.4s)
ProductManager
CoreCompile (7.8s)

```

Step 11: End the sonar Analysis:

```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS COMMENTS
PS D:\Code Quality\ProductManager> dotnet sonarscanner end /d:sonar.token="sqp_93b74155d82503260f39c234d302235c1e0fedda"
SonarScanner for MSBuild 10.3
Using the .NET Core version of the Scanner for MSBuild
Post-processing started.
15:34:33.967 Starting with Scanner for .NET v8 the way the 'sonar.projectBaseDir' property is automatically detected has changed and this has an impact on the files that are analyzed and other properties that are resolved relative to it like 'sonar.exclusions' and 'sonar.test.exclusions'. If you would like to customize the behavior, please set the 'sonar.projectBaseDir' property to point to a directory that contains all the source code you want to analyze. The path may be relative (to the directory from which the analysis was started) or absolute.
Calling the SonarScanner CLI...
INFO: Scanner configuration file: C:\Users\jeswanth.s\dotnet\tools\store\dotnet-sonarscanner\10.3.0\dotnet-sonarscanner\10.3.0\tools\netcoreapp3.1\any\sonar-scanner-5.0.2.4997\bin\..\conf\sonar-scanner.properties
INFO: Project root configuration file: D:\Code Quality\ProductManager\sonarqube\out\sonar-project.properties
INFO: SonarScanner 5.0.2.4997

```

Sonar Report:

Before:

SonarQube community

Projects Issues Rules Quality Profiles Quality Gates Administration More

Embedded database should be used for evaluation purposes only. It doesn't support scaling, upgrading to a new SonarQube Server version, or migration to another database engine. [Learn more](#)

SonarAssessment / main

Overview Issues Security Hotspots Measures Code Activity Project Settings Project Information

Quality Gate ✖ Last analysis 2 minutes ago

Passed

The last analysis has warnings. [See details](#)

New Code Overall Code

Security 0 Open issues A	Reliability 7 Open issues C	Maintainability 9 Open issues A
Accepted issues 0 Valid issues that were not fixed	Coverage 0.0% On 83 lines to cover.	Duplications 7.6% On 4 lines.

After:

SonarQube community

Projects Issues Rules Quality Profiles Quality Gates Administration More

Embedded database should be used for evaluation purposes only. It doesn't support scaling, upgrading to a new SonarQube Server version, or migration to another database engine. [Learn more](#)

SonarAssessment / main

Overview Issues Security Hotspots Measures Code Activity Project Settings Project Information

Quality Gate ✖

Passed

The last analysis has warnings. [See details](#)

New Code Overall Code

Security 0 Open issues A	Reliability 0 Open issues A	Maintainability 2 Open issues A
Accepted issues 0 Valid issues that were not fixed	Coverage 0.0% On 83 lines to cover.	Duplications 7.6% On 1k lines.

Code Review:

Ref No.	Source[Code File] Position (Code Line)	SCORE [1-5]	MEETS STANDARDS	OPPORTUNITIES FOR IMPROVEMENT	MINOR CONCERNS	MAJOR CONCERNS	NOTES	RECOMMENDATIONS	STATUS MARK RESOLVED
1	Models/ModelContext.cs	3	Avoid Warning Message	Yes	Yes	No	In the line 21	remove Warning Message	Resolved
2	Pages/Error.cshtml	2	unread private Field	Yes	Yes	No	in the line 15	Remove the unread Private Field '_Logger'	Resolved
3	Pages/Index.cshtml	2	unread private Field	Yes	Yes	No	in line 8	Remove the unread Private Field '_Logger'	Resolved
4	Pages/Privacy.cshtml	2	unread private Field	Yes	Yes	No	In Line 8	Remove the unread Private Field '_Logger'	Resolved
5	In the Controller folder ProductController.cs file add summary in the code	4	Add summary	Yes	Yes	No	Add summary	/// <summary> /// Get all details /// </summary> /// <returns></returns>	Resolved
6	In the Data folder ModelContext.cs file remove the unused using.	4	Remove Unused using	Yes	Yes	No	Remove the unused using	using System.Collections.Generic;	Resolved
7	In the model folder the Product.cs file, the class name are in the Pascal case.	5	Used Pascal case	Yes	Yes	No	Change the Format	-	Resolved
8	In the model folder the Product.cs file, the namespace are in the correct format.	5	Used Pascal case	Yes	Yes	No	Change the Format	-	Resolved
9	In the model folder the Product.cs file, the entites are in the correct format.	5	Used Pascal case	Yes	Yes	No	Change the Format	-	Resolved
10	In the Data folder ModelContext.cs file the class name are in the Pascal case.	5	Used Pascal case	Yes	Yes	No	Change the Format	-	Resolved
11	In the Data folder ModelContext.cs file the namespace are in the correct format.	5	Used Pascal case	Yes	Yes	No	Change the Format	-	Resolved
12	In the Controller folder ProductController.cs file the private in the correct format.	5	Used_camelCase	Yes	Yes	No	Change the Format	-	Resolved
13	In the Controller folder ProductController.cs file the line number 29 change the get	3	-	Yes	Yes	No	Change the Get to GetById	In the line number 29 the get name change to GetById	Pending